

1. Introduction

NBA player statistics are widely used to compare roles and performance, but many box-score metrics are strongly correlated, so it is often more informative to study the overall structure of a player's statistical profile rather than any single number. In this project, I use 2023–24 NBA per-game player statistics (NBA.com, 2024) to examine (i) which core statistics co-move most strongly across players, and (ii) whether player profiles can be summarized by a small number of interpretable underlying dimensions. After filtering players with sufficient games and minutes for per-game measures to be meaningful, I analyze a core set of variables covering scoring volume, shot mix (including three-point profile), efficiency, playmaking, rebounding, defense, and turnovers. I first describe dependence patterns using the correlation structure (Speagle, 2026a), then apply correlation-based PCA on z-scored variables to obtain a low-dimensional “role map” and interpret the leading components in basketball terms such as offensive involvement and interior–perimeter style. Finally, because PCA relies on second-moment (covariance/correlation) structure and can be sensitive to extreme observations, I include a brief multivariate normality check and squared Mahalanobis distance (D^2) outlier screening (Speagle, 2026b, 2026c) to assess whether unusually distinctive profiles could affect covariance-based summaries and interpretation. The overall structure of this report follows the standard narrative arc for statistical communication (Speagle, 2026a).

2. Data and preprocessing

2.1 Data source and unit of analysis

The dataset and variable definitions are provided in the course Project 1 folder, including `nba_2023_24_stats.csv` and `nba_README.pdf` (Speagle, 2026). This dataset contains **per-game** statistics for NBA players in the 2023–24 season (NBA.com, 2024). Each row represents **one player**, and each number is an **average per game** (for example, “PTS = 20” means the player scores about 20 points per game on average). The raw file includes 572 players and 67 columns (some are rankings and metadata). To make the per-game averages more reliable (so we are not dominated by players with very limited playing time), I follow the dataset recommendation (Speagle, 2026) and keep only players who meet both conditions: **GP** ≥ 10 (played at least 10 games), and **MIN** ≥ 10 (played at least 10 minutes per game). After filtering, the analysis uses **n = 421** players.

2.2 Variables used (18 core statistics, explained in plain language)

I focus on 18 core per-game variables. Below, I explain what each one means in basketball terms:

Playing time / workload

- **MIN**: minutes played per game (how much the player is on the court).

Scoring volume and shot attempts

- **PTS**: points per game (overall scoring output).
- **FGM**: field goals made per game (how many total shots the player makes, including 2s and 3s).
- **FGA**: field goals attempted per game (how many total shots the player takes).
- **FTM**: free throws made per game (how many free throws the player makes).
- **FTA**: free throws attempted per game (how many free throws the player takes).
- **TOV**: turnovers per game (how often the player loses the ball or makes a ball-handling/passing mistake).

Three-point shooting (shot type / style)

- **FG3M**: three-pointers made per game (made 3s).

- **FG3A**: three-pointers attempted per game (attempted 3s; this reflects how “3-point heavy” a player is).
- **FG3_PCT**: three-point percentage (made 3s divided by attempted 3s; a measure of 3-point accuracy).

Shooting efficiency (accuracy)

- **FG_PCT**: overall field goal percentage (FGM/FGA; overall shot-making accuracy).
- **FT_PCT**: free throw percentage (FTM/FTA; free throw accuracy).

Rebounding, playmaking, and defense

- **OREB**: offensive rebounds per game (rebounds collected on the offensive end; often linked to size/positioning/inside play).
- **DREB**: defensive rebounds per game (rebounds collected on defense).
- **REB**: total rebounds per game (OREB + DREB).
- **AST**: assists per game (passes that directly lead to a made basket; a playmaking indicator).
- **STL**: steals per game (taking the ball from the opponent; defensive activity).
- **BLK**: blocks per game (blocking opponent shots; often related to rim protection).

Note on percentage variables: Some players have **0 attempts** in a category (for example, a player might have $FG3A = 0$). In those cases, the percentage value in the dataset may reflect an encoding choice rather than a mathematically meaningful rate. I keep these values to avoid dropping certain player types (especially interior players who rarely shoot threes), but I interpret percentage variables with this limitation in mind.

2.3 Descriptive context (what the typical player looks like)

This filtered sample contains a wide range of player roles—from low-usage bench players to high-usage stars, and from interior bigs to perimeter shooters.

For example: **MIN**: median ≈ 22.3 (mean ≈ 22.6); **PTS**: median ≈ 8.5 (mean ≈ 10.5); **FG3A**: median ≈ 3.0 (range 0.0 to 11.8); **AST**: median ≈ 1.7 (max 10.9)

These wide ranges suggest strong heterogeneity in player roles and motivate a low-dimensional summary (e.g., PCA).

3. Exploratory dependence structure

3.1 Key correlation patterns

To summarize dependence across all 18 variables, I use the correlation heatmap in Figure 1 (Speagle, 2026a). Several patterns stand out:

- **Scoring/usage block:** As expected, scoring is tightly linked to shot volume: $PTS-FGM \approx 0.99$ and $PTS-FGA \approx 0.98$, while minutes also track shot attempts ($MIN-FGA \approx 0.88$). Turnovers rise with usage-related variables as well (e.g., $TOV-FGA \approx 0.85$, $TOV-PTS \approx 0.85$).
- **Free-throw volume:** Free throws made and attempted are nearly identical in correlation terms ($FTM-FTA \approx 0.99$), and both increase with scoring/usage (e.g., $PTS-FTM \approx 0.91$, $PTS-FTA \approx 0.89$).
- **Three-point volume:** Three-point makes and attempts form another tight pair ($FG3M-FG3A \approx 0.99$). Both are moderately related to overall usage (e.g., $FGA-FG3A \approx 0.73$), indicating that higher-usage players often take more threes, but not always.
- **Rebounding structure:** Defensive rebounding and total rebounding are strongly linked ($DREB-REB \approx 0.97$), and offensive rebounds also contribute substantially ($OREB-REB \approx 0.84$).

- Interior vs perimeter tension: Perimeter three-point emphasis tends to oppose interior indicators. For example, $FG3A-FG_PCT \approx -0.38$ and $FG3_PCT-OREB \approx -0.41$, while interior finishing and rebounding align positively ($OREB-FG_PCT \approx 0.62$ and $OREB-REB \approx 0.84$). This contrast already suggests an “interior vs perimeter” stylistic axis that PCA should recover.

Overall, the correlation blocks suggest that a small number of latent dimensions (usage and style/role) explain a large share of player-to-player variation, motivating PCA as a low-dimensional summary of the dependence structure.

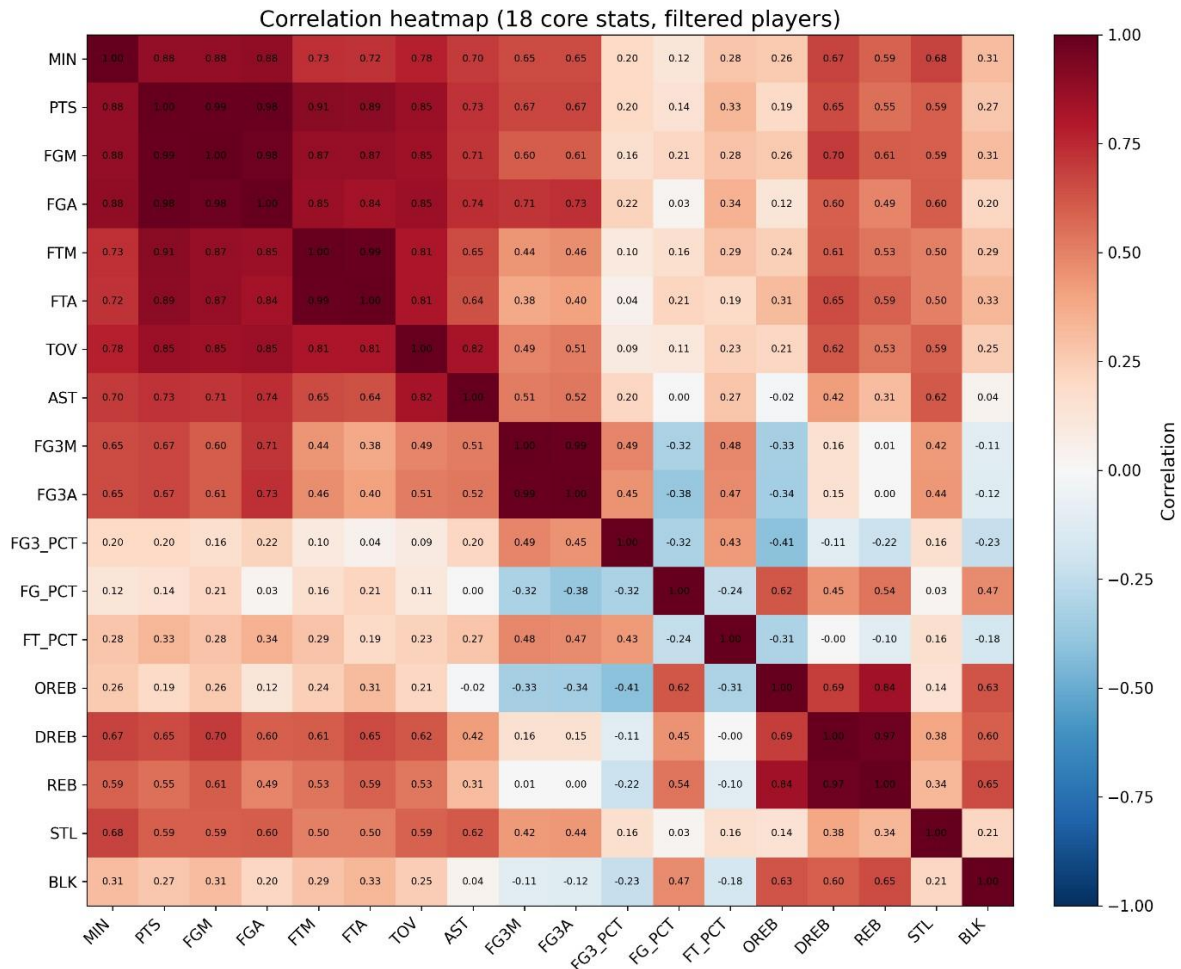


Figure 1. Correlation heatmap of 18 core per-game NBA statistics for filtered players ($GP \geq 10$, $MIN \geq 10$; $n = 421$). Cells show Pearson correlations (annotated to two decimals), with the color scale ranging from -1 (blue) to 1 (red). The heatmap highlights a strongly positively correlated block among scoring/usage variables (MIN, PTS, FGM, FGA, FTM, FTA, TOV, AST), a tight three-point volume pair (FG3M–FG3A), and a rebounding cluster (OREB/DREB/REB), motivating a correlation-based PCA summary in later sections.

4. Methods (PCA + MVN/outlier screening)

This section defines the preprocessing and PCA setup so the subsequent figures and interpretations are reproducible and comparable.

4.1 Analysis plan

Following the correlation-based EDA in Section 3, I then use standardized PCA to identify low-dimensional structure and interpret the leading components using scree plots, loadings, and a PC1–PC2 score map (Section 5). As a “going deeper” step, I assess multivariate normality (MVN) and screen for multivariate outliers using Mahalanobis D^2 and Mardia’s tests (Section 6).

4.2 Standardized (correlation-based) PCA

Because the variables mix counts (e.g., FGA, REB), minutes (MIN), and percentages (FG_PCT, FT_PCT, FG3_PCT), I z-score each variable before PCA, which is equivalent to performing PCA on the correlation matrix (i.e., eigen-decomposing the correlation matrix) (Speagle, 2026c, 2026d). Under this choice, the reported correlation loadings can be interpreted as correlations between each standardized variable and the PC scores (Speagle, 2026d), which makes directions and magnitudes easier to explain. I then use a scree plot to assess how many components capture most variation, use loadings to identify the main drivers of each component, and use a PC1–PC2 scores plot to visualize and interpret player profiles in a two-dimensional space.

4.3 MVN assessment and outlier screening

Because PCA is based on sample covariance structure, unusually extreme observations can affect interpretation. Under an MVN approximation, squared Mahalanobis distances D^2 should be comparable to a χ^2_p reference distribution (Speagle, 2026b); I therefore use a **Mahalanobis D^2 Q–Q plot** to diagnose tail behavior and flag extreme profiles using the high threshold $\chi^2_{0.999,18}$ (Speagle, 2026c). I also report **Mardia’s skewness and kurtosis** tests as formal MVN diagnostics (Speagle, 2026b). Any MVN violations are treated as informative (e.g., heavy tails or mixtures of distinct playing roles) rather than as data errors and are discussed as part of interpretation rather than “fixed” by forcing normality. With the preprocessing and PCA specification fixed, the next section reports the variance explained and motivates a working dimensionality (k) for interpretation.

5. PCA results and interpretation

5.1 Strength of low-dimensional structure

The scree plot (Figure 2) shows very strong low-dimensional structure: PC1 explains 51.2% of standardized variance, PC2 explains 22.6%, and PC1–PC2 together explain 73.8%. This indicates that two dimensions capture a large share of variation in the 18-variable profiles, making a 2D interpretation both meaningful and efficient. For visualization and interpretation, I therefore focus on $k = 2$ as a low-dimensional summary; this is also consistent with the Kaiser criterion for standardized PCA (retain components with eigenvalue > 1) (Speagle, 2026d), since only the first two components clear that threshold while later components contribute comparatively little variance.

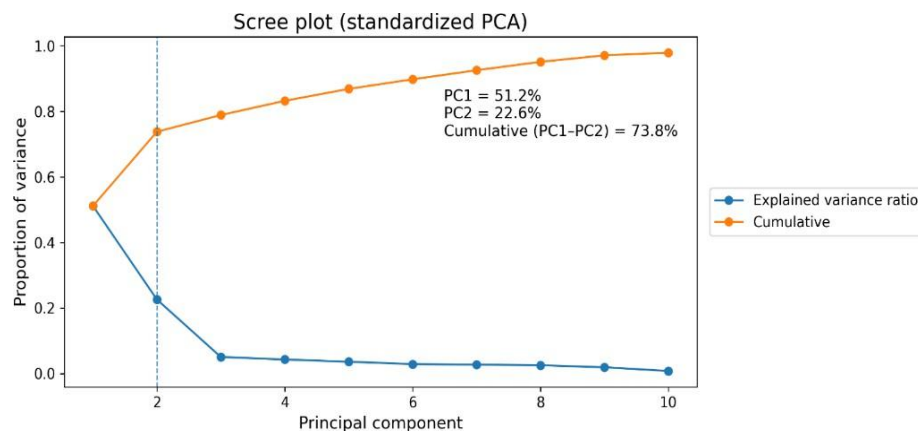


Figure 2. Scree plot for standardized (correlation-based) PCA on the 18 core per-game NBA variables ($n = 421$). The blue curve shows the proportion of variance explained by each principal component and the orange curve shows the cumulative proportion. Variables were z-scored before PCA so that differences in units (counts, minutes, percentages) do not dominate the decomposition. PC1 explains 51.2% of the standardized variance and PC2 explains 22.6%, so the first two components capture 73.8% in total. The sharp drop after PC2 and the dashed vertical line at $k = 2$ indicate strong low-dimensional structure, motivating a two-dimensional representation (PC1–PC2) for interpretation in later sections.

5.2 Interpreting PC1 and PC2 using loadings

I interpret components using the variables with the largest absolute loadings (correlations with the PCs under standardized, correlation-based PCA). Reported values come directly from the PCA loading output and are rounded to three decimals. Because the sign of a principal component is arbitrary, I focus on the largest absolute loadings to identify the main drivers, and use the signs only to describe which variables move together versus in opposition along each axis.

PC1 (usage / offensive involvement): PC1 is driven by scoring and workload measures—PTS (0.979), FGM (0.972), FGA (0.962), and MIN (0.922)—together with other high-involvement signals such as TOV (0.895), FTM (0.894), FTA (0.887), AST (0.773), and DREB (0.737).

Interpretation: PC1 increases with minutes, shot attempts, points, free-throw volume, assists, and turnovers. This pattern is consistent with a “usage / offensive involvement” axis: players with high PC1 tend to be on the court more and carry larger offensive responsibility: they take more shots, score more, draw more free throws, and also accumulate more assists and turnovers as a by-product of higher on-ball involvement. Low PC1 corresponds to limited offensive involvement and smaller roles.

PC2 (interior vs perimeter style): PC2 contrasts interior indicators—OREB (0.867), FG_PCT (0.730), REB (0.688), BLK (0.654), DREB (0.543)—against perimeter three-point emphasis and accuracy—FG3A (−0.650), FG3M (−0.642), FG3_PCT (−0.616), and FT_PCT (−0.534).

Interpretation: PC2 contrasts interior finishing / rim protection / rebounding (positive side) against perimeter three-point emphasis and accuracy (negative side). High PC2 aligns with “big/inside” profiles (offensive rebounding, efficient finishing, rim protection), while low PC2 aligns with perimeter-oriented profiles characterized by higher three-point volume and effectiveness.

5.3 Player “role map” in PC space

Figure 3 plots each player’s scores on (PC1, PC2); since PC1–PC2 explain 73.8% of the standardized variance, the plot provides a compact two-dimensional summary of player-to-player variation in the 18 core statistics. The grey crosshairs mark the origin (average profile under standardization), and the labeled orange points highlight a small set of extreme players to anchor interpretation. Three patterns are most salient:

- **PC1 (left–right)** acts as an **offensive involvement / usage** axis: players on the right have higher overall involvement (minutes and high-volume offensive actions), while players on the left have lower involvement profiles.
- **PC2 (down–up)** acts as a **style/role** axis: higher PC2 corresponds to more interior-oriented profiles (rebounding/blocks and higher inside finishing), whereas lower PC2 corresponds to perimeter-oriented profiles (higher three-point emphasis).
- The broad spread along both axes shows that players vary jointly in “how involved” and “how they play,” supporting the loading-based interpretations in Section 5.2.

This motivates a brief MVN and outlier check to assess whether extreme profiles could influence the covariance/correlation structure underlying the PCA.

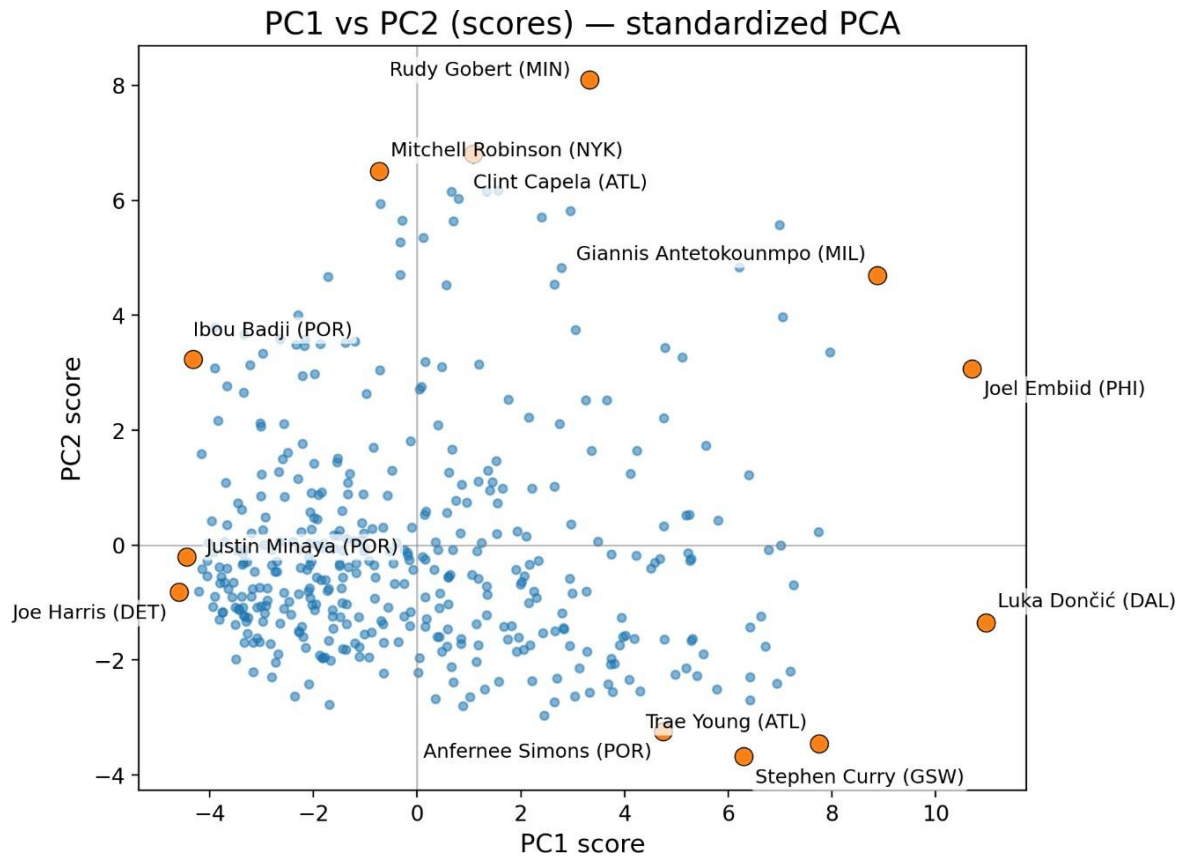


Figure 3. PC1–PC2 score plot from standardized (correlation-based) PCA on 18 core per-game NBA variables ($n = 421$). Each point is one player projected onto the first two components after z-scoring all variables; grey crosshairs mark the origin. Labeled orange points indicate a small set of extreme PC scores (top/bottom on PC1 or PC2) used to anchor interpretation of the two axes as a low-dimensional “role map.” The wide spread along both axes indicates substantial variation in both offensive involvement (PC1) and interior–perimeter style (PC2) across players.

6. Going deeper: MVN assessment and outlier screening

Accordingly, I use Mardia’s tests and a Mahalanobis D^2 Q–Q plot to evaluate multivariate normality and screen for multivariate outliers (Speagle, 2026b).

6.1 Why MVN and outliers matter

MVN assumptions often simplify multivariate inference, but even without formal inference, MVN checks are useful here because PCA is covariance/correlation-based and a small number of extreme player profiles can disproportionately affect the sample covariance and therefore the principal directions. Importantly, flagged observations are not treated as “bad data” by default; in this context they often represent genuine role extremes. The main implication is interpretive: clear non-MVN/heavy tails suggest caution with procedures that rely on MVN for formal inference, while PCA remains a useful descriptive summary of low-dimensional structure (Speagle, 2026c).

6.2 Mardia's tests: strong evidence against MVN

Mardia's multivariate skewness and kurtosis tests on the standardized 18-variable matrix strongly reject MVN: Skewness: $\chi^2 \approx 8640.68$ with $df = 1140$, $p < 1 \times 10^{-16}$ and Kurtosis: $z \approx 71.85$, $p < 1 \times 10^{-16}$. This is not surprising for NBA data: players belong to heterogeneous role groups (guards, wings, bigs, specialists), creating mixture distributions and heavy tails.

6.3 Mahalanobis distance Q–Q plot and flagged outliers

Under MVN, squared Mahalanobis distances D^2 should approximately follow χ^2 with $df = p = 18$. Figure 4 compares empirical D^2 quantiles to χ^2 reference quantiles. The plot shows strong upward deviation in the upper tail, indicating heavy tails and multivariate outliers.

Using a stringent threshold at the 0.999 quantile of χ^2_{18} (≈ 42.31), **23 players** are flagged as extreme. The largest distances include profiles such as **Giannis Antetokounmpo**, **Victor Wembanyama**, and **Luke Kornet**. Importantly, “outlier” here does not necessarily mean data error; it often reflects meaningful extremes in playing style or statistical role. The practical implication is that PCA interpretation should acknowledge that the dataset is not a single homogeneous MVN cloud, and that extreme role profiles exist and may influence covariance-based summaries.

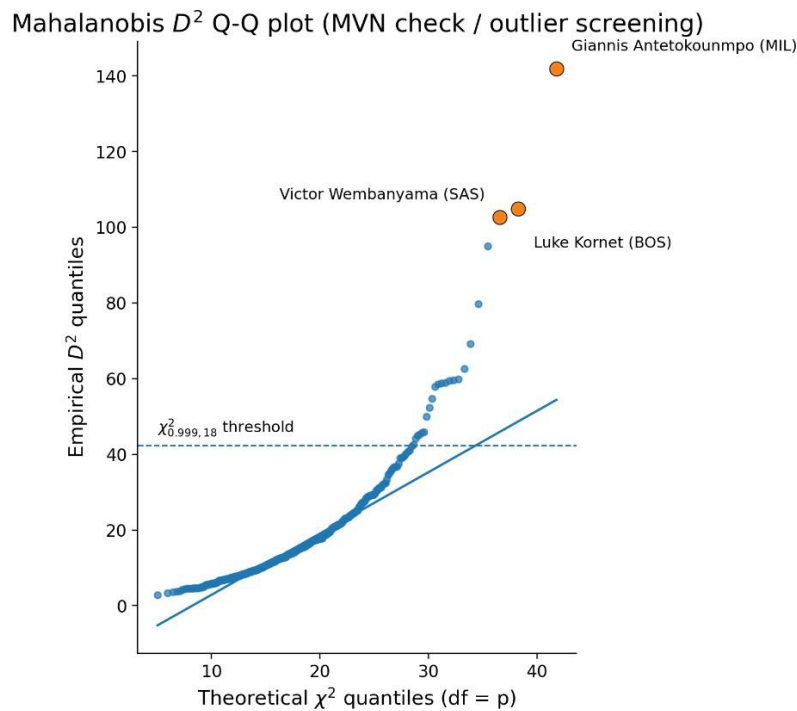


Figure 4. Mahalanobis D^2 Q–Q plot for multivariate normality (MVN) assessment and outlier screening ($n = 421$, $p = 18$). Squared Mahalanobis distances D^2 were computed from the standardized PCA representation by scaling (dividing) each PC score by $\sqrt{\lambda_j}$ (so that, under an MVN model, the transformed scores are approximately $N(0, I)$ and $D^2 \sim \chi^2_p$). The points compare empirical D^2 quantiles to theoretical χ^2 quantiles ($df = p$), and the solid line is a reference line fit using the middle 10%–90% of quantiles for robustness. The dashed horizontal line marks the $\chi^2_{0.999,18}$ threshold used to flag extreme observations; highlighted orange points label the three largest D^2 values as examples. The strong upward deviation in

the upper tail indicates heavy tails and multivariate outliers, consistent with player profiles forming a mixture of distinct roles rather than a single MVN population.

These diagnostics support treating the PC1–PC2 role map as a descriptive summary and interpreting extreme profiles as meaningful role outliers rather than data errors.

7. Discussion and conclusion

This section synthesizes the earlier correlation patterns, PCA results (including the PC1–PC2 role map), and MVN/outlier diagnostics, and summarizes key implications and remaining uncertainties.

This project shows that 2023–24 NBA per-game player statistics have a clear low-dimensional structure. Using correlation-based PCA on 18 standardized variables, PC1 and PC2 explain 73.8% of the total standardized variance, so two axes capture a large share of variation in player season profiles.

The leading components also admit natural basketball interpretations. PC1 reflects usage / offensive involvement: it is driven by minutes and offensive workload measures (points, shot volume, free throws), together with related high-involvement actions such as assists and turnovers. PC2 reflects interior vs perimeter style, contrasting rebounding/blocks/high FG% and offensive rebounding against three-point volume and accuracy, separating “inside big” profiles from perimeter shooting profiles.

As a “going deeper” check, MVN diagnostics (Mardia’s tests and the Mahalanobis D^2 Q–Q plot) provide strong evidence against multivariate normality and reveal a heavy upper tail with multivariate outliers. Importantly, these flagged extremes are not treated as data errors; in this context they plausibly reflect genuine role extremes and a mixture of distinct NBA role groups, so they motivate caution when interpreting covariance/correlation-based summaries.

Limitations. PCA here is a descriptive linear summary and depends on the chosen feature set, per-game scaling, and standardization; the observed heavy tails suggest that extreme profiles can influence covariance/correlation structure.

Next steps. A natural extension is to (i) compare results to a robustness check (e.g., recomputing PCA after removing flagged extremes) and (ii) repeat the analysis using per-possession or rate-normalized features to check whether the same two-axis “role map” persists.

Overall, the two-axis role map provides an interpretable summary of the dominant dependence structure in the 18 statistics, while the MVN/outlier results motivate robustness and stability checks as natural extensions.

8. AI Attribution and Collaboration Statement

I used ChatGPT to help refine figure captions, assist code writing, and improve clarity of written explanations. All structure design, methodological choices, data analyses, code execution, and interpretation decisions were performed by me, and numerical results reported in the text are taken from my computed outputs.

This project was completed independently. No collaboration involved shared code, shared results, or joint writing beyond general course-level discussion.

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